Figure 1. Extended maritime power formula — two-level architecture

Level 1 architecture (general normalized expression) and Level 2 (disaggregation by component)

LEVEL 1 — GENERAL NORMALIZED EXPRESSION

$$MP_{norm} = [\ (IM \times PN) \times SW \times MS^{\gamma} \times CP^{\alpha}\ ]^{1/(n_{eff})}$$

$$n_{eff} = 3 + \gamma + \alpha \cdot MP_{norm} \in (0,\ 1] \cdot \text{All variables} \in (0,\ 1]$$

Preserves MP = MI × NP (Solís Oyarzún, 1997; Uribe Cáceres et al., 2016) · Normalization: geometric mean — UNDP (2010); OECD (2008) · Common-scale anchoring: Colombia → $MP_{norm} \approx 0.53$, same scale as Vargas Suárez et al. (2021, p. 295); the contribution is decomposing it

LEVEL 2 — COMPONENT DISAGGREGATION

MI

Maritime Interests

Political, economic, and security benefits the State obtains from the sea.

$$MI = \Sigma\ w_i \cdot MI_i$$

$$\Sigma w_i = 1;\ w_i \in (0,1);\ MI \in (0,1]$$

Weighted sub-index
Colombia: n = 18 national MI

SOURCES

Till (2007)
Uribe Cáceres et al. (2016)
CCO (2021); OECD (2008)

NP

Naval Power

Military capacity to use, protect, and control the maritime spaces.

$$NP = F \times P \times MDA^{\mu}$$

$$NP \in (0,1];\ \mu = \text{MDA calibration}$$

F: force+C2+tactical CD
MDA elevated to structural level

SOURCES

Mahan (1890); Corbett (1911); Castex (1931)
Till (2007); Bebber (2022)
Uribe Cáceres et al. (2016); Cabuya-Padilla (2024)

SW

Strategic Will

National determination to project and sustain the maritime effort.

$$SW = (PW^{w_1} \cdot IW^{w_2} \cdot SocW^{w_3})^{1/\Sigma w}$$

$$SW \in (0,1];\ \Sigma w_i = 1$$

Weighted geometric mean
MC transversal to PW·IW·SocW

SOURCES

Uribe Cáceres et al. (2016); Till (2007)
Alba-Rocha et al. (2022); Mora Peña & Uribe (2020)
Albarán Gómez (2019); UNDP (2010); Realpe Díaz (2024)

MS

Maritime Security

Degree of control over threats and risks in the maritime domain.

$$MS = (IC^{a_1} \cdot TE^{a_2} \cdot BC^{a_3})^{1/\Sigma a}$$

$$MS \in (0,1];\ \Sigma a_i = 1;\ \gamma = \text{global MS weight}$$

IC: interstate conflicts and gray zone
TE: terrorism · BC: blue crime

SOURCES

Bueger & Edmunds (2024)
IMO (2017; 2021)
Vargas Suárez et al. (2021)

CP

Cyber Power — central contribution of the article

Capacity to operate, defend, and project power in maritime cyberspace.

$$CP = \min\{1,\ (\beta_1 \cdot CS_m + \beta_2 \cdot CD_m + \beta_3 \cdot CR_m) \cdot (1 + \delta \cdot CDip)\}$$

$$CP \in (0,1];\ \Sigma\beta_i = 1;\ \alpha(SCMI) < 1 \rightarrow \text{concave: initial returns} \uparrow$$

CS_m: cyber security · CD_m: cyber defense
CR_m: cyber resilience · CDip: cyber diplomacy

SOURCES

Nye (2010); Bebber (2022); Cano (2024)
Cabuya-Padilla (2024, 2025)
Realpe Díaz (2024); Caro Bejarano (2013)

TRANSVERSAL METHODOLOGICAL DECISIONS

MDA: not an independent variable — distributed as an enabling factor across NP(F), MS, and CP (Bebber, 2022; Bueger & Edmunds, 2024)

Maritime Consciousness (Uribe Cáceres et al., 2016, p. 64): a cultural-identity construct, distinct from technical-operational MDA — transversal to PW·IW·SocW in SW

CDip (Realpe Díaz, 2024, pp. 297–299): dual role — projection coefficient in CP *and* expression of PW in SW, without analytical duplication. Model of an analytical-conceptual nature, not computational (Solís Oyarzún, 1997; Till, 2007)